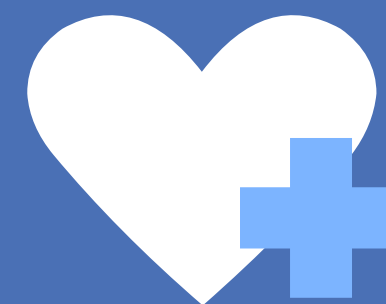




DECISION SUPPORT SYSTEM FOR HEART DISEASE RISK PREDICTION



GROUP 5

presented for : Miss Nurul Nadzirah binti Mohd Hasri



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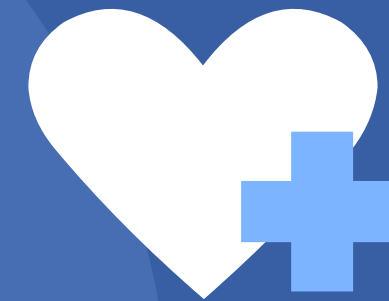


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Introduction



- Heart disease remains one of the leading causes of death worldwide
- Early detection of heart disease risk is crucial to prevent serious complications
- Manual analysis of multiple patient medical indicators is time-consuming and error-prone
- A Decision Support System (DSS) can analyse patient data efficiently and consistently to support healthcare professionals in making faster and more accurate clinical decisions.
- This project develops a data-driven and model-driven DSS using medical data to predict heart disease risk
- The prediction results are presented through a Power BI dashboard to support effective clinical decision-making



Problem Identification

Problem 1

Late detection of high risk heart disease patients

Problem 3

High workload due to a large number of patients and limited healthcare professionals

Problem 2

Manual and time consuming analysis of patient screening

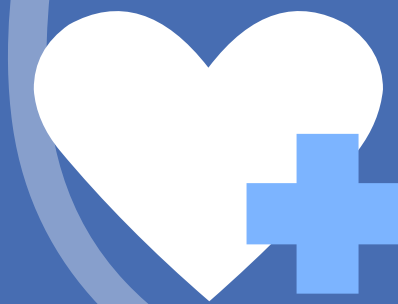
Problem 4

Increased treatment cost .

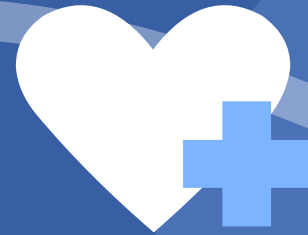


Objective

- 1** To develop a Decision Support System (DSS) for predicting heart disease risk based on medical data from the patient.
- 2** To develop a simple and user-friendly dashboard system that can help healthcare professionals visualise the risk level of patients and key medical indicators.
- 3** To minimize manual analysis and enhance the accuracy of patient screening with the help of model-driven decision support.
- 4** To support clinical decision making by classifying patients as low, moderate and high risk levels of heart disease.



User Requirements



Primary Users

- Doctors (Cardiology / General Physicians)
- Medical Staff
- Hospital Management

User Requirements

- Doctors need clear patient risk categorisation (low, Moderate, High) to support clinical screening decisions.
- Medical staff require simple visual indicators to quickly interpret patient health status
- Hospital Management requires aggregated statistics to monitor heart disease trends and resource allocation



Implementation Strategy

System Development

- Patient medical data is collected from existing healthcare databases
- The data is cleaned and processed to ensure accuracy and consistency
- Analytical and predictive models are applied to classify patients into different heart disease risk level
- The processed results are integrated into a power bi dashboard

System Usage

- Healthcare professionals access the dashboard to review patient risks level
- High risk patient are prioritised for further screening and medical intervention



Scope

- * The system focuses on developing a data-driven and model-driven Decision Support System (DSS) for heart disease risk prediction.
- * The scope includes data cleaning, model development, risk classification, and dashboard visualisation.
- * The DSS uses secondary medical data and presents insights through a Power BI dashboard.
- * The system supports clinical screening decisions but does not replace medical diagnosis.

Significance

- Enables early identification of high-risk heart disease patients.
- * Reduces manual workload for healthcare professionals.
 - * Supports faster and more accurate clinical decision-making.
 - * Helps healthcare organisations plan preventive strategies and allocate medical resources effectively.



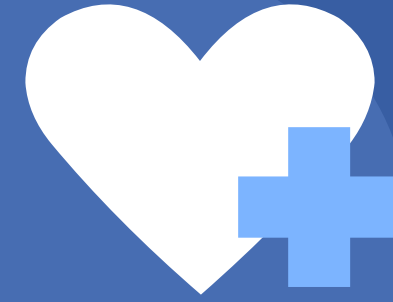
Literature Review

Existing Approaches

- Researchers have applied machine learning models for heart disease prediction (Liza et al., 2025)
- Ensemble models often outperform traditional statistical methods (Prasetyo et al., 2023)
- Fuzzy logic-based DSS improves interpretability of medical decisions (Kumar et al., 2022)
- Data-driven systems help analyse complex and large-scale medical data (Miao et al., 2025)



Literature Review



Research Gap & Justification

- Many systems focus only on prediction accuracy without clear visualisation
- Limited emphasis on user-friendly dashboards for healthcare professionals
- Need for integrated systems that combine:
 - Predictive models
 - Risk categorisation
 - Visual analytics

(prasetyo et al., 2023 ; Kumar et al., 2022 ; Miao et al., 2025; Liza et al., 2025)

Based on the literature, this project focuses on a data-driven and model-driven DSS that integrates predictive models with visual analytics to support medical decision making



Methodology

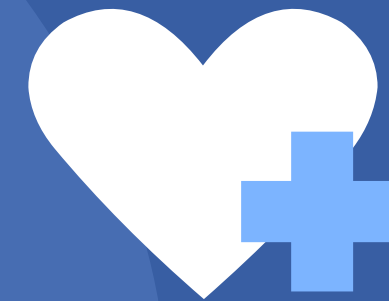
- Model-driven Decision Support System (DSS)
- Analytical models as the core decision element
- Combines: (Statistical analysis & Machine learning models)
- Transforms raw data into actionable insights

Methodology Phase

- Preliminary Study
- Data Collection (Secondary Dataset)
- Data Dictionary Preparation
- Model Development & Analysis
- System Usability Testing
- System Development & Dashboard Integration

DSS Components:

- Data Management : Patient medical dataset
- Model Management : Statistical and machine learning models
- User Interface : Power BI Dashboard for decision support



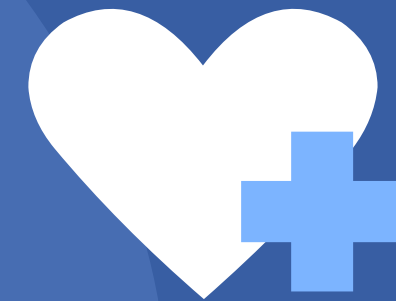
Preliminary Study

Purpose of Preliminary Study

- To understand the heart disease problem domain
- To identify key medical indicators related to heart disease risk
- To justify the use of a data-driven & model-driven DSS

Activities Conducted

- Reviewed past healthcare research and clinical literature.
- Identified common risk factors such as age, cholesterol, and blood pressure.
- Evaluated dataset suitability for predictive modelling tasks
- This phase ensures that only relevant medical indicators are selected to build reliable decision models within the DSS.



Data Acquisition

Data Source

Secondary data from Kaggle (Heart Disease Dataset).

Data Source

Dataset: Heart Disease Dataset

Key Attributed Collected

- Age
- Sex
- Chest pain type
- Resting blood pressure
- Cholesterol level
- Fasting blood sugar
- ECG results
- Maximum heart rate
- Heart disease (target variable)

This dataset is suitable for a decision Support System as it contains structured clinical attributes commonly used by doctors during cardiovascular risk assessment.



System Usability

1. Cards



Key Indicators

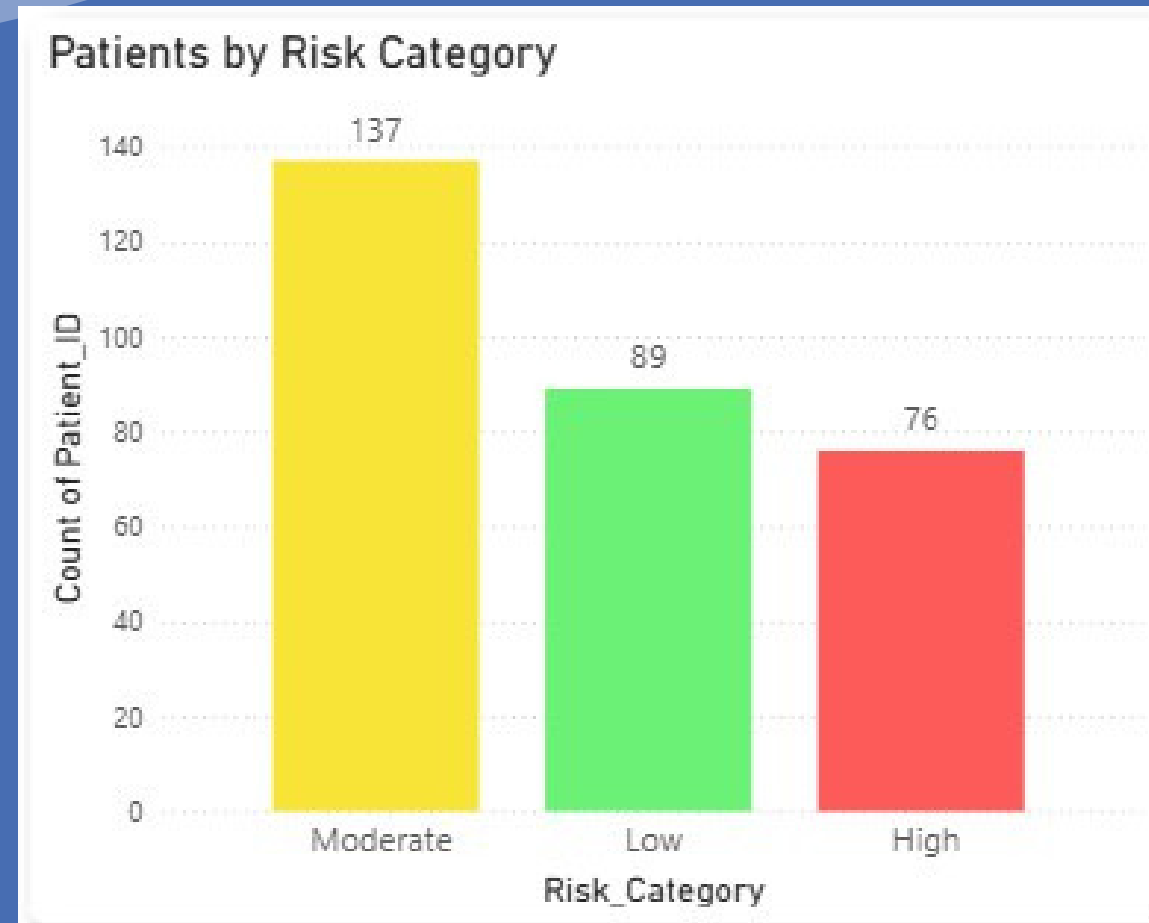
- Total Patients: total number of patient records.
- High Risk Patients: number of patients classified as high risk (priority for screening).
- High Risk Rate: percentage of high-risk patients from total population.
- Heart Disease Cases: number of confirmed heart disease cases (Heart_Disease = 1) for validation.
- Average Risk Score (High Risk): indicates the severity level of high-risk patients.

Kpi Cards

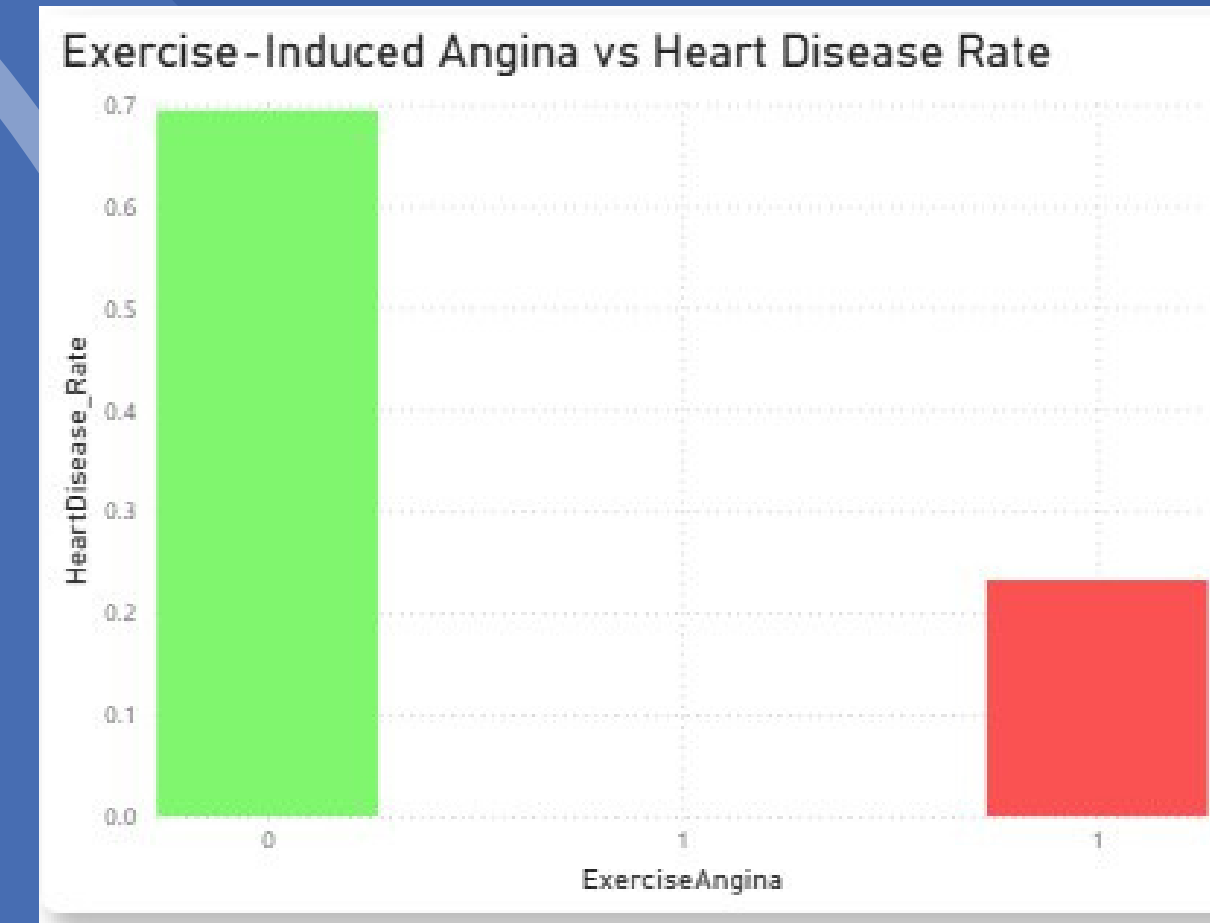
- Provides a quick overview of key heart disease risk screening statistics for medical professionals.
- Helps users understand the dataset at a glance before exploring detailed charts.
- These Kpi cards help healthcare professionals quickly identify the scale of high-risk patients and prioritise clinicals screening decisions.



2. Clustered Column Chart

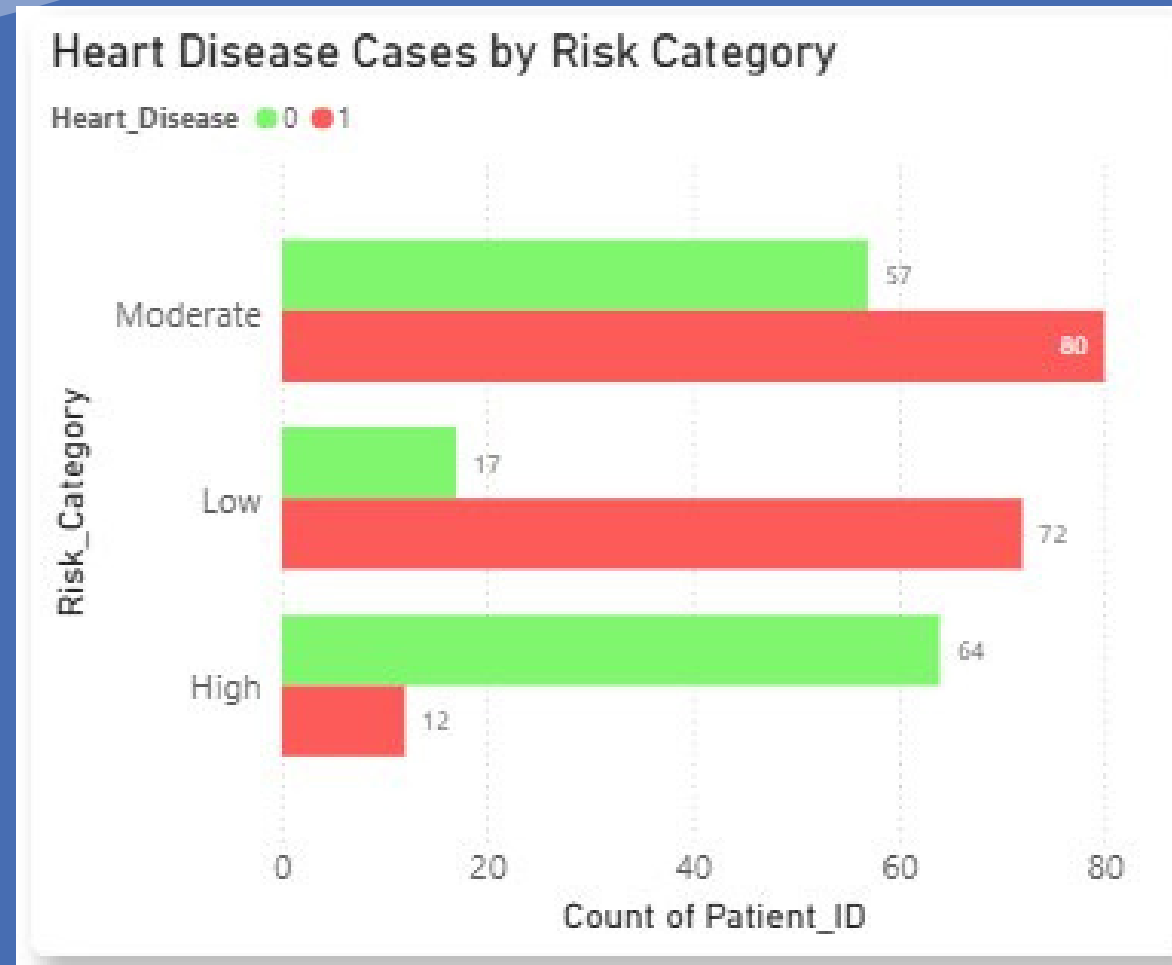


- Shows the distribution of patients across three risk groups: Low, Moderate, High.
- X-axis: Risk Category
- Y-axis: Number of patients
- The chart indicates Moderate risk has the most patients, followed by Low, while High risk is the smallest group.
- The visualization supports doctors in prioritising moderate and high risk patients for further cardiovascular screening

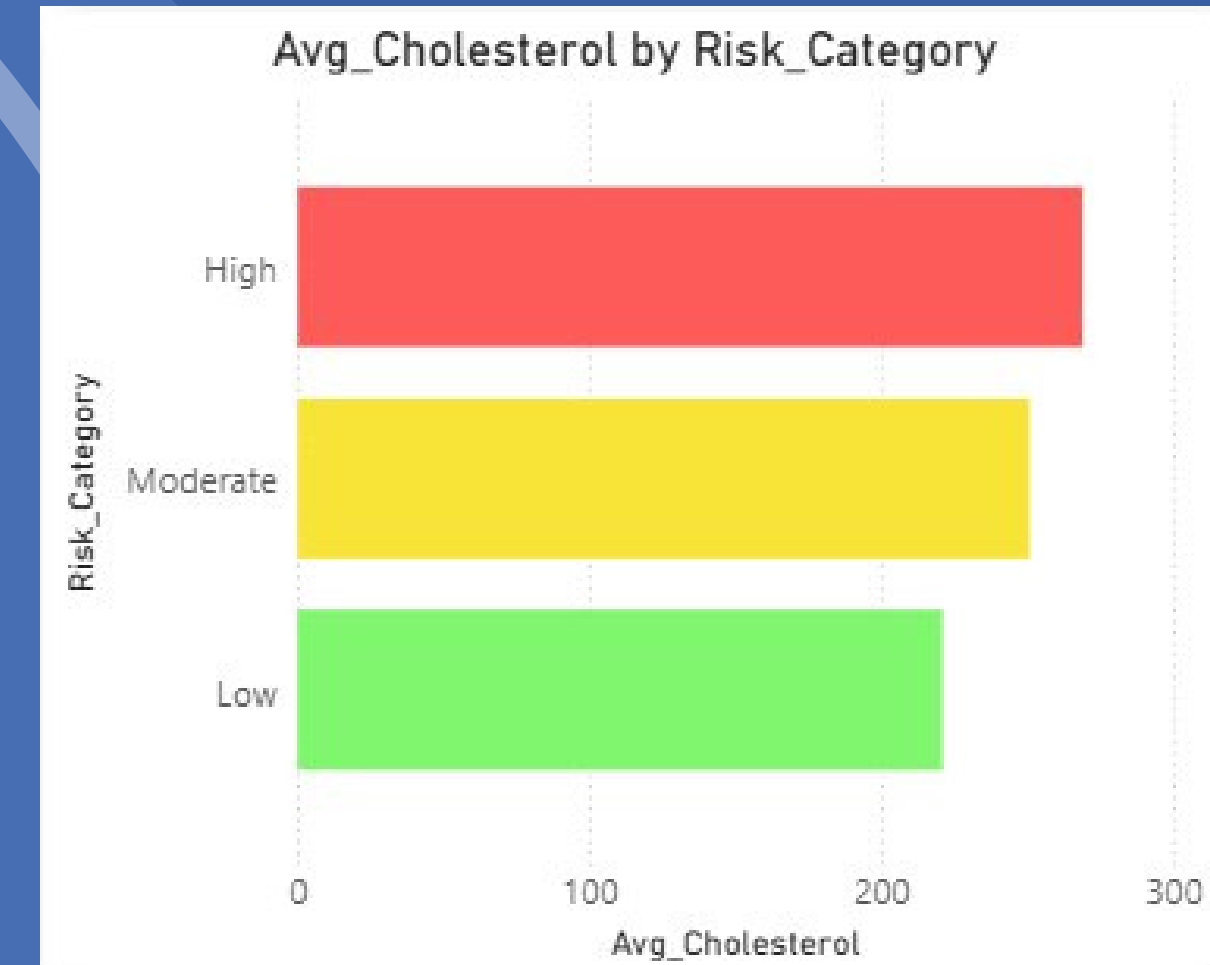


- Compares heart disease rate between patients with and without exercise-induced angina.
- X-axis: Exercise-induced angina (0 = No, 1 = Yes)
- Y-axis: Heart disease rate (proportion of disease cases)
- Helps identify whether exercise-induced angina is a strong symptom indicator of heart disease.
- Supports clinical decision-making by highlighting patients who may require further cardiovascular screening.

3. Clustered Bar Chart

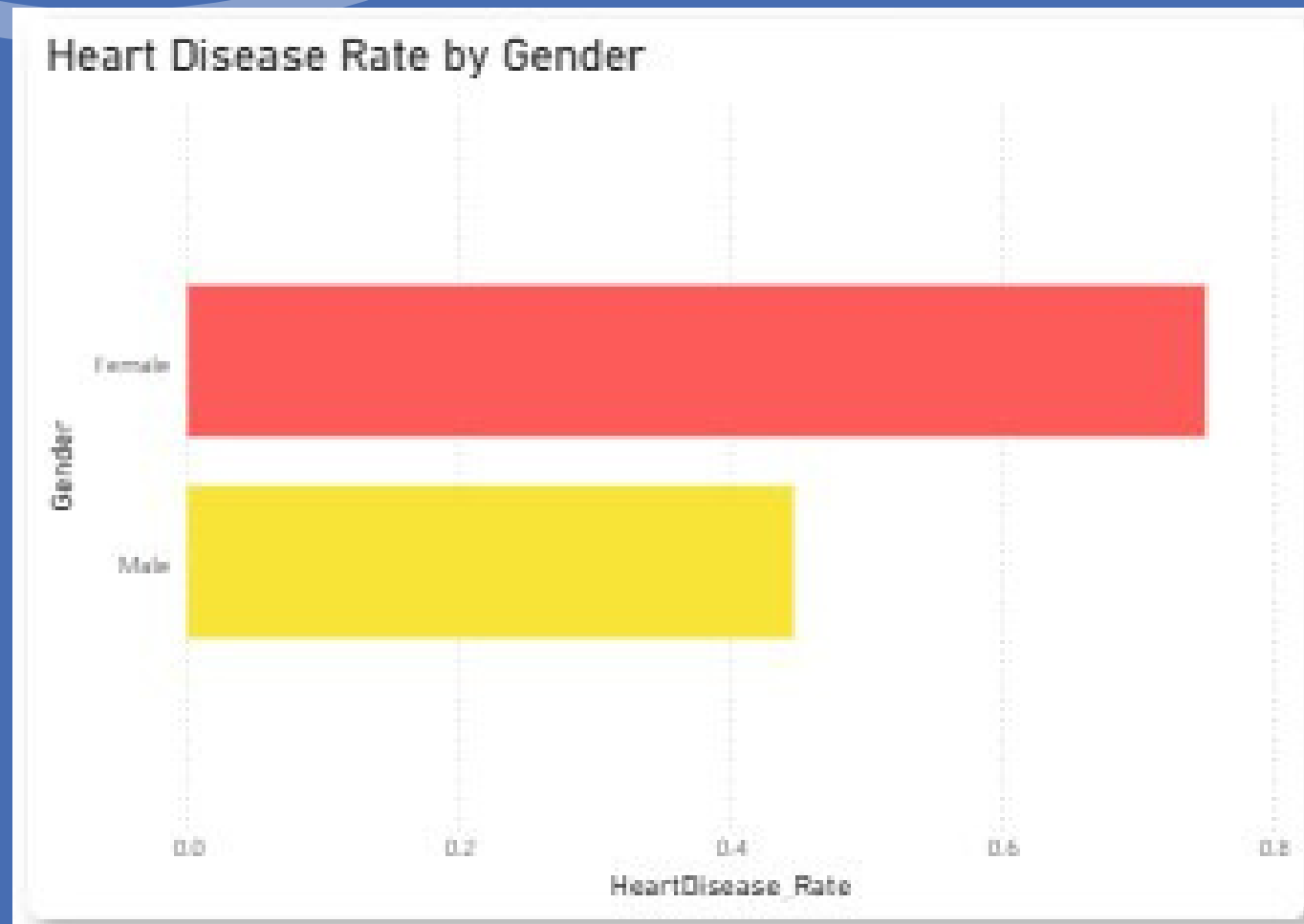


- Compares actual heart disease outcomes across predicted risk groups: Low, Moderate, High.
- Y-axis: Risk Category
- X-axis: Number of patients
- Legend: Heart Disease (0 = No disease, 1 = Disease)
- Used to validate the risk classification, showing whether higher risk groups contain more confirmed heart disease cases.
- This chart validates the DSS risk classification and increases confidence in using the system for early screening decisions



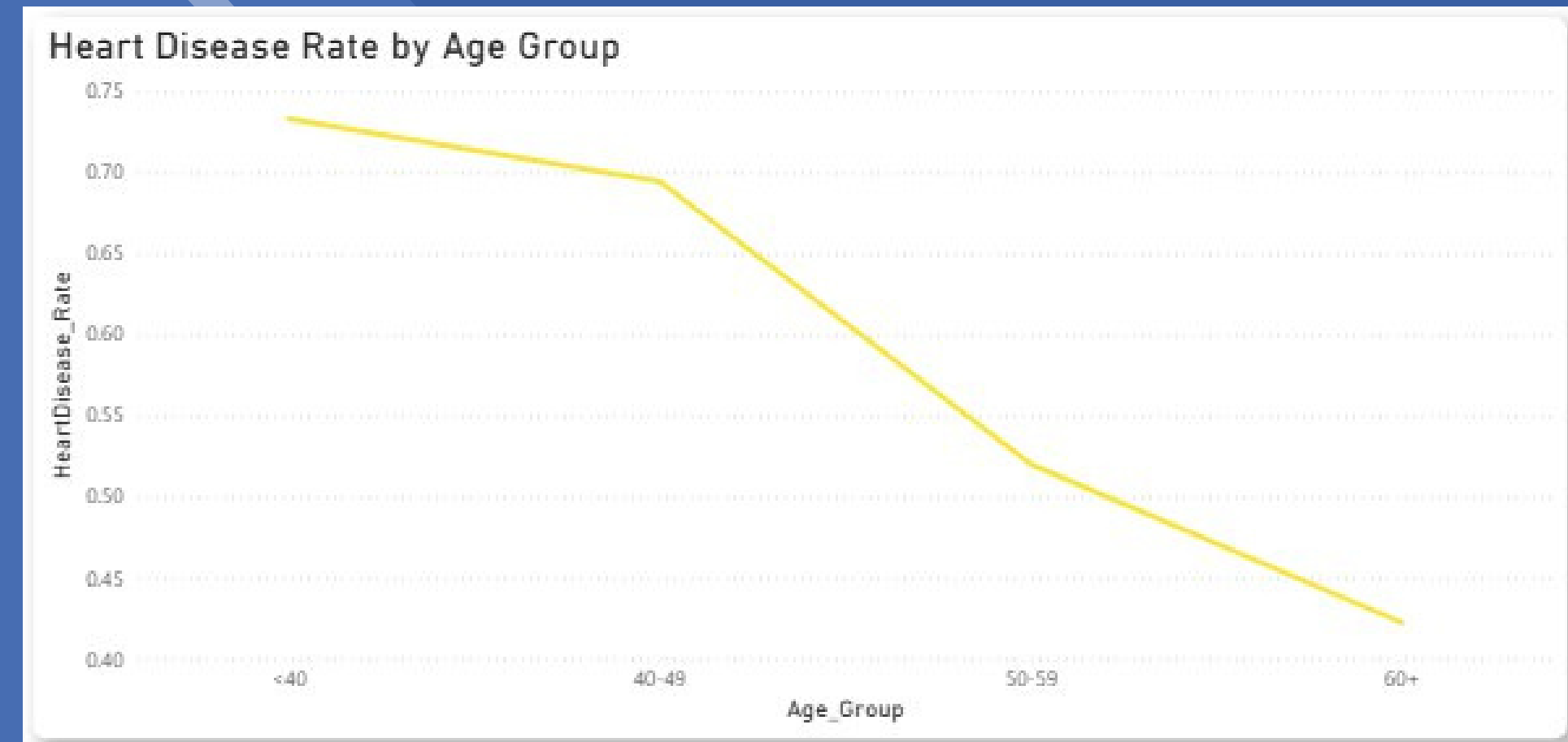
- Shows the mean cholesterol level for each risk group: Low, Moderate, High.
- Y-axis: Risk Category
- X-axis: Average cholesterol value
- Helps explain why patients are classified into different risk levels.
- The chart shows High-risk patients have the highest average cholesterol, supporting the clinical link between high cholesterol and heart disease risk.

4.Cluster Bar Chart



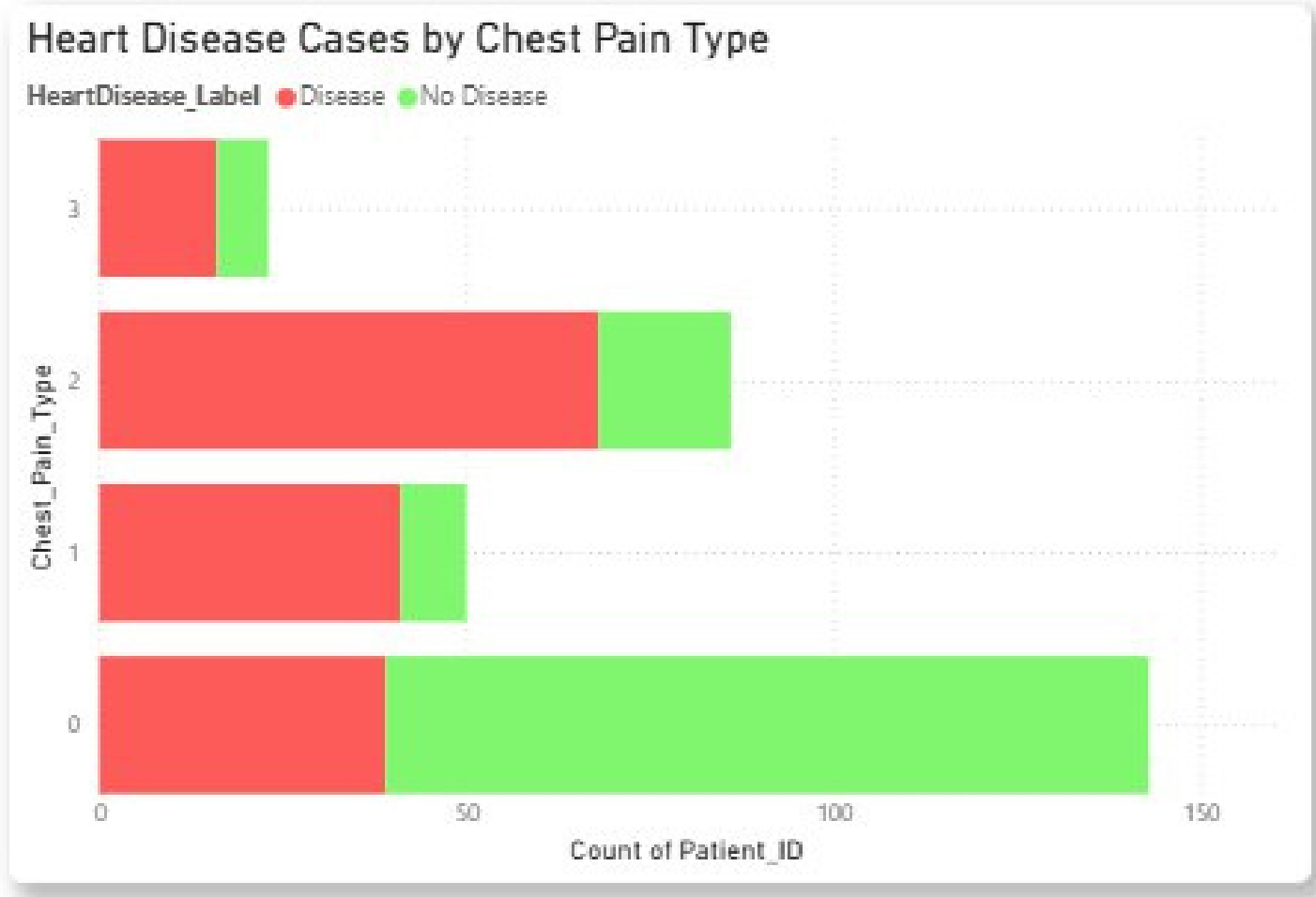
- Shows the comparison of heart disease rate between Male and Female patients.
- X-axis: Heart disease rate
- Y-axis: Gender
- Helps identify which gender has a higher prevalence of heart disease in the dataset.
- Supports demographic-based screening by highlighting gender groups that may require more attention and early monitoring.
- This visualisation supports decision support screening by helping healthcare professionals prioritise patients based on gender-related risk trends.

5.Line Chart



- Shows the trend of heart disease rate across age groups: <40, 40–49, 50–59, 60+.
- X-axis: Age Group
- Y-axis: Heart disease rate (% of patients with disease)
- Helps identify which age groups have higher prevalence, supporting demographic-based screening.
- Highlights that risk increases with age, which aligns with clinical knowledge and supports early detection planning.

6.Stacked Bar Chart



- Shows the relationship between chest pain type and heart disease occurrence.
- Y-axis: Chest Pain Type
- X-axis: Number of patients
- Stacked by: Disease vs No Disease
- Helps identify which chest pain categories have a higher proportion of heart disease cases.
- Clinically important because chest pain is a major symptom used in cardiovascular screening, supporting early detection and risk assessment.

7.Table

High-Risk Patients List						
Patient_ID	Age	Gender	Chest_Pain_Type	ExerciseAngina	Risk_Score	Risk_Category
22	55	Female		0	1	10 High
23	56	Female		0	1	10 High
26	59	Male		0	1	10 High
27	60	Male		0	1	10 High
37	70	Male		2	1	10 High
18	51	Male		0	1	9 High
22	55	Male		0	1	9 High
24	57	Male		0	1	9 High
25	58	Female		0	1	9 High
28	61	Female		0	1	9 High
28	61	Male		2	1	9 High
29	62	Female		0	1	9 High
34	67	Male		0	1	9 High
35	68	Male		2	1	9 High
13	46	Male		0	1	8 High
18	51	Male		0	1	8 High
20	53	Male		0	1	8 High
22	55	Male		0	1	8 High
23	56	Female		0	1	8 High

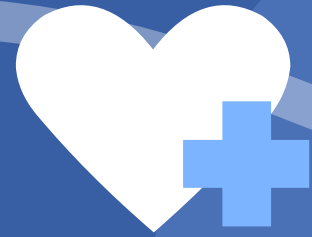
- Displays the list of patients classified as High Risk for prioritised screening.
- Columns include: Patient ID, Age, Gender, Chest Pain Type, Exercise Angina, Risk Score, Risk Category.
- Helps medical professionals quickly identify which patients need immediate follow-up.
- Risk Score allows ranking patients by severity to support clinical decision-making.
- These table outputs assist healthcare professionals in prioritising high-risk patients for timely and evidence-based screening decisions.

Interface



- Accessible layout for quick interpretation by healthcare professionals
- The interface is designed to minimise cognitive load and allow quick interpretation during clinical consultations.

System Benefits & Measurements



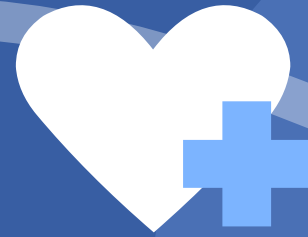
System Benefits

- Enables early detection of high risk heart disease patients
- Reduces manual workload for healthcare professionals
- Supports faster and more accurate clinical decision-making
- Improves prioritisation of patients for screening and intervention

Measurement Scheme

- Prediction accuracy of the risk classification model
- Reduction in manual screening time
- User satisfaction and ease of use of the dashboard

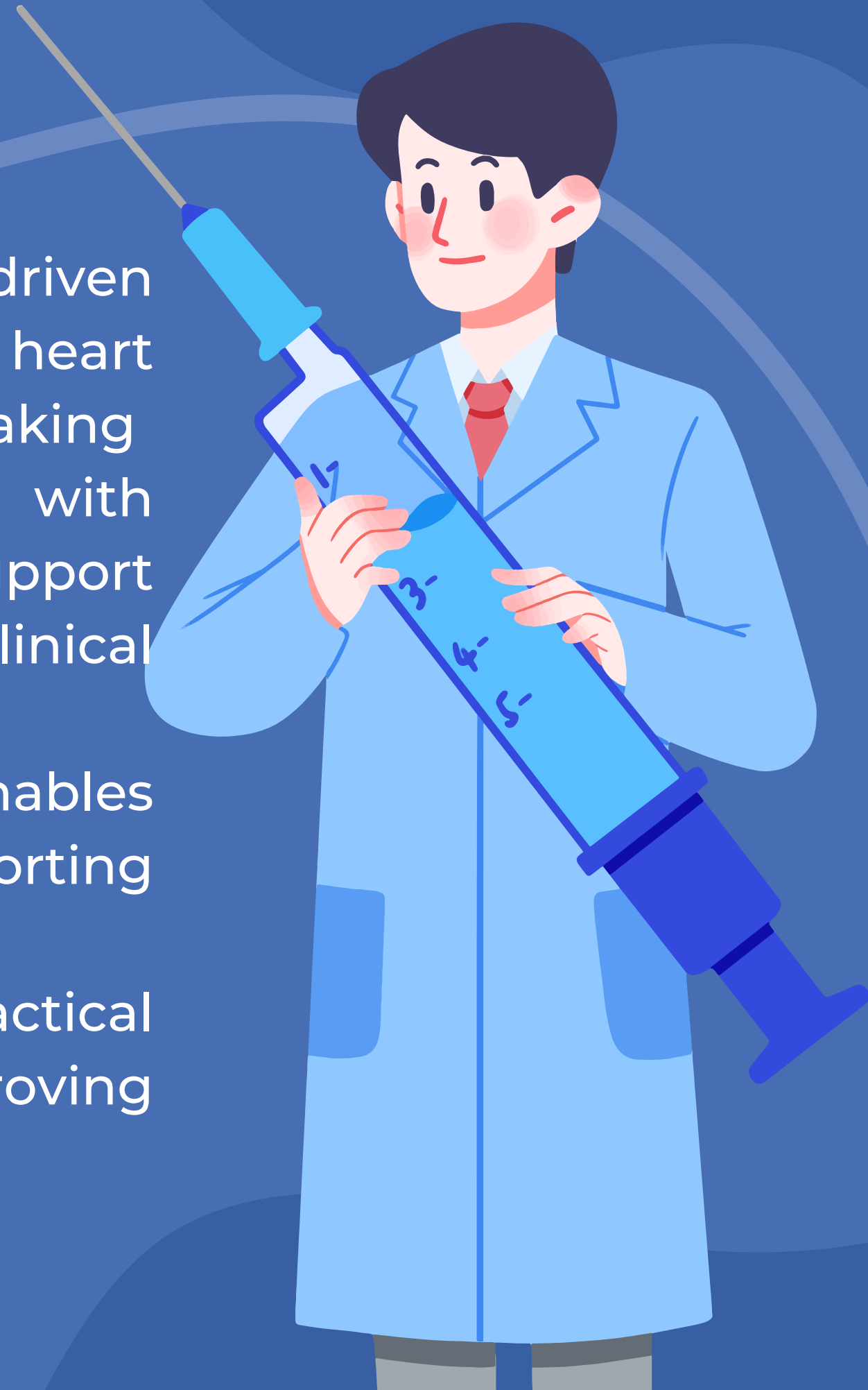
Future Work



- Expand dataset with larger and more diverse patient records to improve model generalizability.
- Integrate additional medical indicators (e.g., lifestyle factors, genetic markers) for richer risk prediction.
- Enhance explainability features to provide clearer reasoning behind DSS recommendations.
- Explore real-time deployment in clinical settings for faster decision support.

Conclusion

- This project successfully developed a data driven and model-driven DSS strategy to support heart disease risk prediction and clinical decision making
- The DSS integrates patients medical data with analytical models and visuals dashboard to support faster, more accurate and evidence based clinical decision-making.
- the system reduces manual workload and enables early identification of high risk patients, supporting preventive healthcare planning
- Overall, the proposed DSS demonstrates the practical value of decision support system in improving healthcare outcome



THANK
YOU ♥ +

